

PRANAV UPRETI

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EDUCATION

University of Toronto

Toronto, ON

Engineering Science, Engineering Physics (intended major)

2024 – 2029

- **Relevant Courses** : Process Modelling and Simulation (audited), ESC190 - Data Structures and Algorithms, CSC2332 - Introduction to Quantum Computing (audited)
- **Skills**: MATLAB, C/C++, Python (NumPy, SciPy, Keras, TensorFlow, PennyLane, Qiskit, FastAPI, Streamlit), Java, SystemVerilog, RISC-V, gdsfactory/KLayout, Tidy3D, Git, uv/pytest, HTML/CSS/JS

EXPERIENCE

Undergraduate Researcher

Feb. 2026 – Present

Joyce Poon Group, University of Toronto

Toronto, ON

- Built a **benchmarking suite** for PhIDO, our **agentic photonic integrated circuit (PIC)** design tool. It tests an agentic framework's ability to reproduce novel PICs from papers and foundry data sheets.
- Added functionality to toggle PhIDO's PDK retrieval, knowledge-base retrieval, physics checkers, and formal verification.
- Implemented **GDSFactory+** deterministic routing engines into PhIDO's layout agent. This transitioned sloppy design to near fab-ready layouts.

Research Volunteer, Ballistocardiography Signal Analysis

May 2026 – Aug. 2026

Mount Sinai Hospital | Supervisor: Dr. Isaac Sung Jae Chang

Toronto, ON

- Developed **double-integration** code estimating apparent cardiac **stroke volume** from 3-axis **ballistocardiogram (BCG)** signals, recovering hemodynamic markers that heart rate alone cannot capture.
- Validated BCG heart-rate extraction on a healthy cohort and heart-failure patients undergoing right heart catheterization.

Embedded Programmer

November 2025 – Present

University of Toronto Biomedical Engineering Design Team (UTBIOME)

Toronto, ON

- Writing **C++** code for on demand droplet generator used in **medical research**.
- Implementing control logic for a centrifuge motor to control droplet generating through a microcontroller that interfaces with a centrifuge motor.

PROJECTS

plot2spectra | *Python, Computer Vision, Claude API*

- Built a tool that **digitizes published figures back into numerical data**.
- Reimplementation of a paper's optical-flow inspired algorithm which traces curves by minimizing a composite segmentation, intensity, smoothness, and semantic loss over image gradients.
- Replaced hand-tuned heuristics for axis and tick detection with **Claude Vision**; packaged with **uv** and covered by a **pytest** suite.

Virtual Aircraft Controller | *FastAPI, Streamlit, higgs-audio API*

- Developed a real-time, interactive virtual aircraft controller using **FastAPI** for backend and **Streamlit** for front end, complete with audio inputs and aircraft movement visualization
- Integrated speech to text models NLP pipelines via **higgs-audio API** to extract heading, altitude, and runway assignment from speech input.
- Used **Git** for version control and collaboration, with detailed commit history to ensure project reproducibility.

SlideHive | *CAD, Thermal Analysis with SimScale*

- A beehive redesigned for the Toronto Beekeeping Collective to reduce lifting strain by **>50%** and maximize bee health
- Has similar thermal insulation to currently used Langstroth hives, tested using **thermal analysis** in SimScale
- Custom designed drawer mechanism that **maximizes longevity** by reducing propolis buildup (a resin bees produce).